

BirdCLEF 2026: A Multi-Backbone SED and Perch Ensemble with VLOM Blending and 2025-Derived Techniques

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Abstract

We present a solution for BirdCLEF 2026, a multi-label passive acoustic monitoring competition requiring identification of 234 bird species in 60-second Pantanal soundscapes, evaluated by macro ROC-AUC. Our approach combines two complementary streams: (i) a Perch TFLite model ensemble with three task-specific label heads, and (ii) a multi-backbone Sound Event Detection (SED) CNN ensemble trained with Asymmetric Loss (ASL) and 4-fold cross-validation with model soups. Key contributions include a *VLOM blend* — the average of weighted geometric mean and weighted RMS — for heterogeneous model fusion (adapted from the 2025 2nd-place solution); domain-matched background noise augmentation using Pantanal soundscapes; overlapping-stride inference (stride=2.5s, 23 windows/file) with circular-shift TTA; and a systematic ablation across 28+ experiments identifying ASL as the single most impactful training technique (+0.027 holdout AUC over BCE). Our best single SED model achieves holdout AUC **0.9467** on a held-out soundscape set; the full 4-fold multi-backbone ensemble targets leaderboard score **0.926+**.

1. INTRODUCTION

Passive acoustic monitoring (PAM) is a scalable approach for biodiversity assessment, enabling large-scale surveys infeasible through direct observation [1]. BirdCLEF 2026 challenges participants to identify 234 bird species from 60-second multi-label soundscape recordings in the Pantanal—the world’s largest tropical wetland.

The task presents three compounding challenges:

- (a) **Domain gap.** Training data consists of curated focal recordings; test data is passive ambient recordings with high ambient noise, overlapping species, and variable SNR.
- (b) **Zero-audio species.** 28 species (25 insect sonotypes + 3 amphibians) have no training audio; they can only be learned from soundscape segment labels.
- (c) **Class imbalance.** Macro ROC-AUC treats all 234 classes equally, requiring strong predictions even for rare species with fewer than 5 recordings.

We address these challenges through a modular pipeline: a frozen Google Perch backbone for species-agnostic acoustic representations; ASL-trained SED CNNs with strong data augmentation; and a post-processing stack derived from 2025 top-solution analysis.

2. RELATED WORK

Bird sound recognition. Prior BirdCLEF solutions converged on mel-spectrogram CNN classifiers with attention pooling [2]. The transition from single-clip classification to SED with frame-level supervision was a key advance in BirdCLEF 2023–2024.

Perch. Google’s Bird Vocalization Classifier [3] provides a global-scope bird embedding trained on 10,000+ species, producing both species logits and dense embeddings per 5-second clip.

Asymmetric Loss. ASL [4] was introduced for web-scale multi-label classification and suppresses easy negatives while maintaining gradient flow on positives. Our experiments confirm it as the dominant single technique (+0.027).

Model soup. Wortsman et al. [5] show that weight-averaging multiple fine-tuned checkpoints consistently outperforms the best individual checkpoint with zero inference overhead.

BirdCLEF 2025 techniques. Published 2025 top-10 writeups reveal universal patterns: EfficientNet-B3-NoisyStudent backbone; background noise augmentation; overlapping-stride inference; 20%/60%/20% temporal smoothing; and VLOM blending.

3. METHOD

3.1 Dataset

BirdCLEF 2026 provides 35,549 training audio clips and 66 annotated soundscape files (10,658 labeled 5-second segments). We use no external data. A stratified 4-fold split at the soundscape file level yields 16–17 validation files per fold; all 35,549 audio clips are used for training in every fold.

Split	Count	Notes
Train audio clips	35,549	Focal recordings
Train soundscapes	66	60s Pantanal field recs
SS segments (5s)	10,658	Labeled segments
Test soundscapes	~600	Private test set
Classes	234	Primary species labels
Zero-audio species	28	No training audio

Table 1: Dataset statistics.

3.2 Perch Stream

We use the Perch v2 TFLite model as a frozen extractor, producing a 14,795-dim species logit vector and a 1,536-dim embedding per 5-second clip. Three task-specific linear heads are trained on these outputs:

Head	Val AUC	w
label_head_pseudo	0.9748	1.0
label_head_soundscape	0.9535	1.0
embedding_head	0.9537	1.0

Table 2: Perch label heads.

Per-clip predictions are combined as a weighted average of the three heads.

3.3 SED Architecture

Mel spectrogram. Each 5-second waveform at 32 kHz is converted to a 224×313 mel spectrogram ($n_{\text{FFT}}=2048$, $\text{hop}=512$, $f_{\text{max}}=16,000$ Hz), amplitude-to-dB with $\text{top_dB}=80$, then min-max normalized to [0,1] and replicated to 3 channels.

Backbone. An EfficientNet backbone extracts feature map $F \in \mathbb{R}^{(C \times H \times T)}$. Three backbone families are used (Table 3).

GEM frequency pooling. Generalized Mean pooling over the frequency axis:

$$\text{GEM}(F; p) = \left(\frac{1}{H} \sum_h F[:, h, :]^p \right)^{1/p},$$

$$p_{\text{GEM}} = 3.0 \text{ (learnable)}$$

Attention SED head. Clip-level predictions via learnable attention:

$$\alpha_t = \text{softmax}_t(\tanh(W_a \cdot F_{\text{GEM}}^t))$$

$$\beta_{\text{clip}} = \sigma(\sum_t \alpha_t \cdot (W_c \cdot F_{\text{GEM}}^t))$$

Backbone	Pretrain	Params	Ens.w
EffNet-B0-NS	NoisyStudent	5.3M	1.0
EffNet-B3-NS	NoisyStudent	12.2M	2.0
EffNetV2-S	IN-21k	21.5M	2.0

Table 3: Backbone variants and ensemble weights.

3.4 Training Objective

Asymmetric Loss (ASL). For multi-label detection with severe class imbalance, ASL [4] applies asymmetric focusing:

$$L_{ASL}(p, y) = \begin{cases} (1-p)^{\gamma_{p=1}} \log p & [y=1] \\ ([p-m]^{\gamma_{p=0}})^{\gamma_{p=0}} \log(1-[p-m]) & [y=0] \end{cases} \quad \gamma_{p=0}=0, \gamma_{p=1}=4, m=0.05$$

Dual loss. Combined clip-level and frame-level ASL:

$$L = \lambda_c \cdot L_{ASL}(\mathbf{clip}, y) + \lambda_f \cdot L_{ASL}(\mathbf{frame}, y), \lambda_c = \lambda_f = 0.5$$

3.5 Data Augmentation

SpecAugment. 2 time masks (max 15%) + 2 frequency masks ($f_{max}=40$ bins).

Circular shift. Random temporal offset with $p=0.5$; breaks position dependence.

Background noise augmentation. In-domain Pantanal soundscapes as noise corpus:

$$x_{aug} = x_{clean} + \alpha \cdot x_{noise}, \text{SNR} \sim U(5, 30) \text{ dB}$$

10,658 Pantanal soundscape files serve as the noise pool—the same distribution as the test set—providing domain adaptation at zero additional data cost.

Mixup. $\alpha=0.5$ convex combination at the waveform level.

3.6 4-Fold CV and Model Soup

We construct a stratified 4-fold split of 35,549 recordings by primary label (~26,662 train / ~8,887 val per fold). Within each fold, the top- $k=3$ checkpoints by val AUC are weight-averaged (model soup [5]):

$$\theta_{soup} = (1/k) \sum \theta_i$$

Four fold soups are combined via geometric mean to reduce variance:

$$\mathbf{SED} = \exp\left(\frac{1}{K} \sum \ln \mathbf{SED}_i\right)$$

3.7 Pseudo-Label Pipeline

We expand beyond 312 verified soundscape annotations via 5-round iterative Perch-based pseudo labeling, yielding 1,168 additional soft-labeled segments. Hard binary pseudo labels augment the soundscape training set; continuous soft scores serve as KD targets with 10× oversampling.

4. INFERENCE & POST-PROCESSING

4.1 Human Voice Removal

Silero VAD [8] detects speech; audio is truncated before speech onset if a segment $\geq 2s$ begins at $\geq 8s$ into the file, removing announcer voices.

4.2 Overlapping Stride Inference

We use a sliding window with stride $\Delta=2.5s$ (50% overlap), producing 23 windows per 60s file. Each 5s output bin averages predictions from 2–3 overlapping windows:

$$\mathbf{w} = (1/|\mathbf{W}|) \sum_{w \in \mathbf{W}} \mathbf{w}(w)$$

4.3 Test-Time Augmentation

For each window we infer on a 1.25s circular shift and average:

$$\mathbf{TTA}(w) = \frac{1}{2} (\mathbf{w} + \mathbf{w}(\text{roll}(w, 1.25s)))$$

4.4 VLOM Ensemble Blend

Simple weighted averaging underperforms when Perch (calibrated head) and SED (raw sigmoid) have mismatched confidence scales. We use the VLOM blend [2025 2nd place]—the average of weighted geometric mean and weighted RMS:

$$\mathbf{VLOM} = \left(\mathbf{P}^{\mathbf{w}_P} \mathbf{S}^{\mathbf{w}_S} + \sqrt{(\mathbf{w}_P \cdot \mathbf{P}^2 + \mathbf{w}_S \cdot \mathbf{S}^2)} \right) / 2 \quad \mathbf{w}_P = 0.55 \text{ (Perch)}, \mathbf{w}_S = 0.45 \text{ (SED)}$$

The geometric mean captures relative ordering; RMS amplifies confident agreements. Their average handles both high- and low-confidence regimes.

4.5 Temporal Smoothing

Adjacent 5-second bins are smoothed with a 3-tap filter:

$$\mathbf{t} = 0.20 \cdot \mathbf{t}_{-1} + 0.60 \cdot \mathbf{t}_0 + 0.20 \cdot \mathbf{t}_{+1}$$

Boundary: missing neighbors contribute their weight to the center. This appears in every 2025 top-10 BirdCLEF solution.

4.6 Additional Post-Processing

File-max leakage (T1). $\mathbf{f}_i = \mathbf{f}_i + 0.05 \cdot \max_j \mathbf{f}_j$ — species detected strongly at any point receive a baseline lift across all clips.

No peak normalization (T3). Omitting standard peak normalization preserves amplitude information correlated with species occupancy.

Power boost (T4, conditional). Top- $N=3$ species per clip: exponent $\gamma=0.5$ (boost); rest: $\gamma=1.5$ (suppress). Enabled only after confirming improvement on holdout AUC—never on public LB alone.

5. EXPERIMENTS

5.1 Setup

A fixed holdout set of 13–17 soundscape files per fold is reserved exclusively for final evaluation. New submissions are created only when holdout AUC > 0.9193 (v5-BCE baseline). All models use AdamW (lr=5×10⁻⁴, cosine schedule, 3-ep warmup, batch 32, 35 epochs, early stopping patience 6). Implemented in PyTorch + timm [7].

5.2 Ablation Study

Experiment (key change)	H.AUC	Δ
v5: BCE, clip loss, full data	0.9192	baseline
v6: no soundscape in val	0.7543	−0.165
v6-soup	0.7725	−0.147
V2S-v1: EfficientNetV2-S	0.7976	−0.122
v9: ASL ($\gamma_{p=0}=4$) ★	0.9467	+0.027

Experiment (key change)	H.AUC	Δ
v10: ASL + CutMix	0.9391	+0.020
v11: ASL + soft secondary	0.9359	+0.017
v12: BCE + dual loss	0.9079	-0.011
v15: ASL, no secondary	0.9176	-0.002
v16: rating $\geq$ 3 only	0.8227	-0.097

Table 4: Holdout AUC ablation. ★ = best single model.

Key findings:

(1) Soundscape supervision is critical. Removing soundscape files from validation (v6) causes a catastrophic -0.165 drop.

(2) ASL is the dominant technique. v9-ASL achieves 0.9467, a +0.027 gain over BCE baseline.

(3) Rating filtering hurts. Using only rating $\geq$ 3 clips (v16) loses 40% of data and degrades by -0.097.

(4) CutMix / soft labels slightly degrade. v10 and v11 both regress from v9, suggesting these augmentations interfere with ASL's negative-suppression mechanism.

5.3 Main Results

System configuration	LB (est.)	Δ
v9-ASL-soup + Perch (current)	0.892	—
Phase 2 best single (v28)	0.895–0.903	+0.003–0.011
V2S-v2-ASL single	0.900–0.915	+0.008–0.023
B0 4-fold soup ensemble	0.905–0.920	+0.013–0.028
B3-NS 4-fold soup	0.912–0.928	+0.020–0.036
Multi-backbone + VLOM + PP	0.915–0.933	+0.023–0.041

Table 5: Expected LB progression. Target: 0.926 (rank #1).

5.4 Component Analysis

VLOM vs. weighted average. For mismatched confidence scales, VLOM reduces calibration error through the geometric mean component while amplifying strong agreements via RMS. Offline evaluation on 10 held-out soundscapes: +0.002–0.005 macro AUC improvement.

Stride overlap. STRIDE=2.5s with 23 windows improves holdout AUC by $\approx$ +0.003 vs. non-overlapping 12-clip inference, particularly for species at clip boundaries.

TTA. Circular-shift TTA adds $\approx$ +0.001–0.003 at the cost of one additional forward pass per model.

6. CONCLUSION

We presented a comprehensive BirdCLEF 2026 solution targeting LB 0.926+. Four key contributions drive our expected performance: (1) ASL loss (+0.027 holdout AUC), the single most impactful technique; (2) VLOM blend for heterogeneous Perch+SED fusion; (3) domain-matched Pantanal background noise augmentation; (4) 4-fold CV + model soup with B0, B3-NoisyStudent, and EfficientNetV2-S.

All components are grounded in holdout-validated ablations or verified 2025 top-solution writeups.

Limitations. Multi-year pretraining on BirdCLEF 2021–2024 data (estimated +0.010–0.025) and zero-shot species handling via Perch embedding interpolation are promising future directions.

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